Workforce Capacity Forecasting System

Fri,Oct 9,2026

Summary

The team presented an AI-supported system for forecasting workload and planning workforce capacity.

- Combines forecasting, capacity analysis, risk detection, and planning.
- Capacity matching uses employee experience and task requirements.
- Forecasting algorithm remains undecided; prediction quality will guide selection.
- The project plan spans 12 weeks.

Action Items

- No action items generated

Key Discussion Points

Project Problem and Motivation

Soujanya introduced the workforce-capacity forecasting project and described the difficulties of managing changing workloads with manual methods. She explained the need to anticipate overload, underutilization, and resource shortages so managers can plan earlier.

Objectives and Research Gap

Vinay presented the project objectives and summarized related research on workload forecasting and resource allocation. Sujan Neelgund described the research gap and the proposed platform's combined forecasting, capacity, risk, and planning functions.

System Workflow and Technical Questions

Sujan Neelgund explained the system's seven-layer workflow, from data input through dashboard reporting. The instructor asked which forecasting and capacity-analysis techniques the team would use, and Sujan said he was not yet sure about the algorithms.

Modules and Capacity Calculation

Tejaswini outlined the system modules, including workforce and task management, preprocessing and forecasting, and capacity and risk analysis. After questions about preprocessing and capacity calculations, Soujanya described matching employee experience to task requirements and clarified that machine learning would not be used for team capacity calculation.

Tools, Deliverables, and Timeline

The team identified Python as its primary programming language and described expected outputs, including forecasting, capacity and risk analysis, recommendations, a dashboard, and a prototype. Vinay presented a 12-week plan and discussed the research and reference materials informing the project.

Conclusion and Presentation Feedback

Vinay summarized the system's intended combination of workload forecasting, capacity analysis, risk detection, and resource planning. The instructor praised the team and encouraged clearer explanations, stronger communication, and greater confidence.